

Design FMEA (qualitative) — T8-40g

VBF-T8R1-DFMEA-01 Case t8_r1 — Long-endurance drone battery 2026-08-25

Prepared: _____ Reviewed: _____ Approved: _____

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- **Document code**: VBF-T8R1-DFMEA-01
- **Case**: t8_r1 — long-endurance drone battery
- **Generation date**: 2026-08-25
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Qualitative version, based on simulation risk signals (values from tool output files under `cell/`). Severity/occurrence are three-level qualitative ratings (high/medium/low), basis = magnitude of simulation value deviating from the threshold; RPN uses the simplified qualitative S×O matrix (annotated qualitative caliber).

Failure Mode Table

Failure mode	Failure cause	Simulation signal (detectability basis)	Severity	Occurrence	RPN (qual.)	Design-side mitigation recommendation
Negative electrode lithium plating (fast charge)	Overpotential at anode during 4C charge exceeds plating threshold; slow solid diffusion (large particles); low anode capacity headroom	`r5_final_4c45_dfn.j son` anode_potential_v min = +0.0082 V (baseline plated; after mitigation margin 8.2 mV)	High	Low	High×Low → acceptable	Fine-graphite 1.5 μm anode particles, $t_{\text{anode}} \geq 0.5$, thicker anode (95 μm) for lithiation headroom; keep 4C charge within voltage-limited acceptance (0.94 Ah)
Thermal runaway risk (temperature rise exceeds limit)	Polarization + ohmic heat at 4C/45 °C exceeds cooling capacity	`r5_final_4c45_dfn.j son` T_max 330.41 K vs 333.15 K limit — margin only 2.74 K	High	Low	High×Low → acceptable	Forced-air cooling $h = 40 \text{ W/m}^2/\text{K}$ assumed; verify real h by thermal test; derate charge current if ambient > 45 °C
Electrolyte oxidative decomposition (voltage window)	Solvent HOMO vs cathode potential at 4.2 V	No DFT endorsement run (real_compute = false; endorse entry skipped) — signal absent	Medium	Medium	Medium×Medium → track	Not endorsed at molecular level; if Stage-5 endorsement is later funded, verify electrolyte HOMO window first
Insufficient capacity	Capacity below drone endurance requirement	1C capacity 6.9427 Ah vs nominal 6.1494 Ah — PASS (energy 25.246 Wh)	Medium	Low	Medium×Low → acceptable	Capacity margin exists (DFN 1C exceeds nominal); monitor on production lots
Mass overrun (>40 g)	Layer mass accumulation from tolerance spread	Contract mass 39.798 g vs 40 g — margin only 0.202 g (0.5%)	Medium	Medium	Medium×Medium → track	Tight mass margin: control foil thickness and coating density tolerances in production; electrolyte mass (11.03 g, literature density) is outside contract caliber but counts in real pack mass
5C retention degradation	Diffusion/transport limits reappear at high rate after aging	5C retention 98.77% (fresh) — PASS; aging not simulated	Medium	Low	Medium×Low → acceptable	Run aging + 5C protocol if cycle-life requirement is added later
Incomplete 4C charge (voltage-limited)	4.2 V cut-off reached at 0.94 Ah during 4C charge	`r5_final_4c45_dfn.j son` capacity 0.9395 Ah	Low	High	Low×High → acceptable	Charge at ≤ 2C for full capacity; 4C reserved for emergency/top-up

Conclusion

- **Highest-risk items**: (1) thermal margin at 4C/45 °C (2.74 K) — mitigation implemented via $h = 40 \text{ W/m}^2/\text{K}$ cooling, but real heat-transfer coefficient must be verified physically; (2) electrolyte oxidation at 4.2 V — no molecular-level endorsement in this case (real_compute = false); (3) mass margin (0.202 g) — production tolerance control required.
- Mitigation measures for plating, thermal, and retention risks have been implemented in the design (R2–R4 change history, see design_spec.md §6).
- Complete FMEA including process/supplier failures: N/A (beyond pure simulation boundary).